

CodeAlpha Matrix Operations Program in C

Matrix Addition and Transpose using 2D Arrays

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter Matrix A:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter Matrix B:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("\nMatrix Addition:\n");
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }

    printf("\nTranspose of Matrix A:\n");
    for(i=0;i<c;i++) {
        for(j=0;j<r;j++) {
            printf("%d ",a[j][i]);
        }
        printf("\n");
    }

    return 0;
}
```